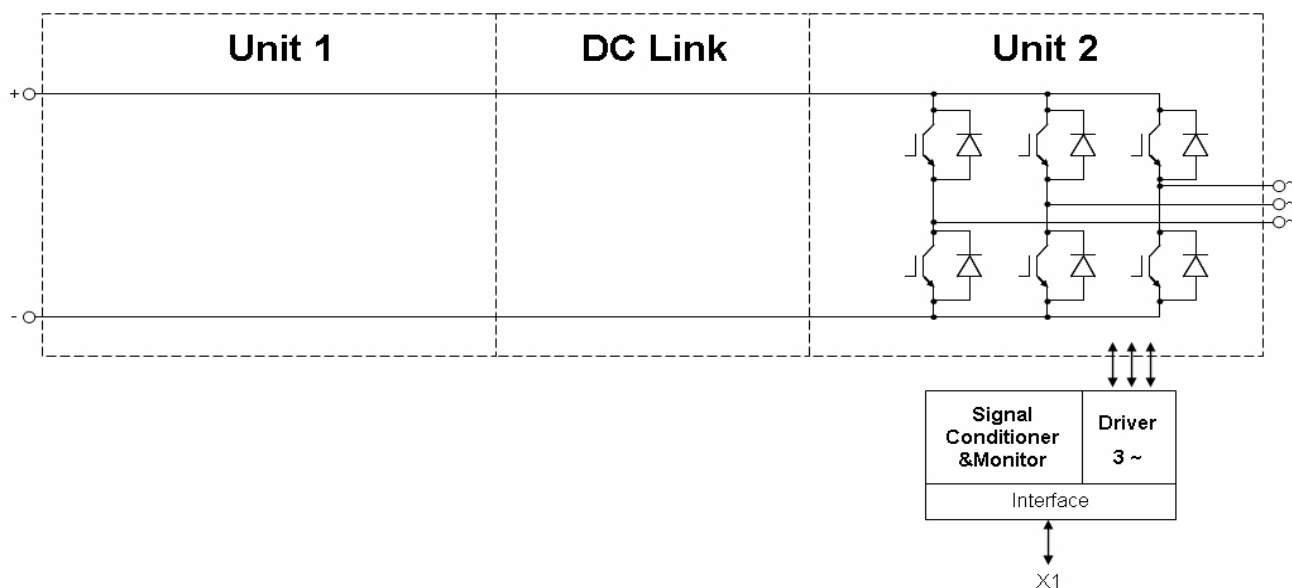
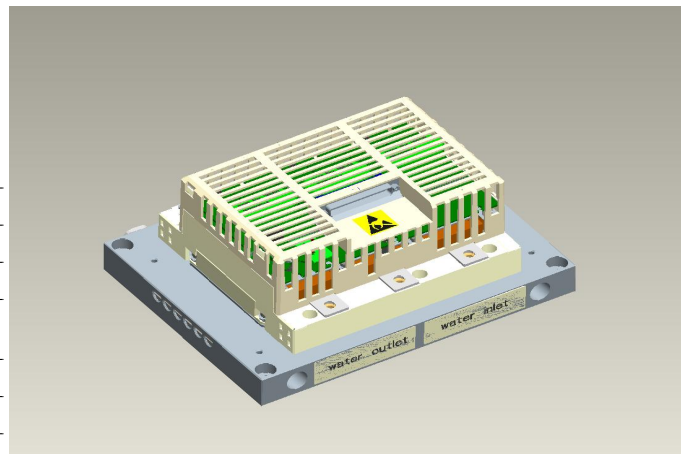


General information

IGBT Stack for typical voltages of up to 500 V_{RMS}
Rated output current 300 A_{RMS}

- High power converter
- Solar power
- Motor drives
- 62mm power module

Topology	B6I
Application	Inverter
Load type	Resistive, inductive
Semiconductor (Inverter Section)	3x FF450R12KE4
Heatsink	Water cooled
Implemented sensors	Current, voltage, temperature
Driver signals IGBT	Electrical
Design standards	UL 94, prepared for UL 508C
Sales - name	6PS04512E43W39693
SP - No.	SP001129256



prepared by: OW	date of publication: 2013-08-05
approved by: AR	revision: 2.0

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Characteristic values

DC Link

			min.	typ.	max.	
Rated voltage		V_{DC}		650		V

Notes

The voltage sensor VM110 is only used for measurement. It is realized no over-voltage shutdown.

Inverter Section

			min.	typ.	max.	
Rated continuous current	$V_{DC} = 800\text{ V}$, $V_{AC} = 500\text{ V}_{RMS}$, $\cos(\varphi) = 0.85$, $f_{AC\ sine} = 5\text{ Hz}$, $f_{sw} = 2500\text{ Hz}$, $T_{inlet} = 40^\circ\text{C}$, $T_j \leq 125^\circ\text{C}$	I_{AC}			300	A_{RMS}
Continuous current at low frequency	$V_{DC} = 800\text{ V}$, $V_{AC} = 500\text{ V}_{RMS}$, $f_{AC\ sine} = 0\text{ Hz}$, $f_{sw} = 2500\text{ Hz}$, $T_{inlet} = 40^\circ\text{C}$, $T_j \leq 125^\circ\text{C}$	$I_{AC\ low}$			220	A_{RMS}
Rated continuous current for 150% overload capability	$I_{AC\ 150\%} = 330\text{ A}_{RMS}$, $t_{on\ over} = 60\text{ s}$, $t_{recovery} = 600\text{ s}$, $T_j \leq 125^\circ\text{C}$	$I_{AC\ over1}$			220	A_{RMS}
Over current shutdown	within 15 μs	$I_{AC\ OC}$		625		A_{peak}
Power losses	$I_{AC} = 500\text{ A}$, $V_{DC} = 800\text{ V}$, $\cos(\varphi) = 0.85$, $f_{AC\ sine} = 5\text{ Hz}$, $f_{sw} = 2500\text{ Hz}$, $T_{inlet} = 40^\circ\text{C}$, $T_j \leq 125^\circ\text{C}$	P_{loss}		2400		W

Controller interface

Driver and interface board	ref. to separate Application Note		DR210			
			min.	typ.	max.	
Auxiliary voltage		V_{aux}	18	24	30	V
Auxiliary power requirement	$V_{aux} = 24\text{ V}$	P_{aux}			40	W
Digital input level	resistor to GND 10 k Ω , capacitor to GND 1 nF, logic high = on	$V_{in\ low}$	0		1.5	V
		$V_{in\ high}$	11		15	V
Digital output level	open collector, logic low = no fault, max. 15 mA	$V_{out\ low}$	0		1.5	V
		$V_{out\ high}$		15		V
Analog current sensor output inverter section	load max 5 mA, @ 300 A_{RMS}	$V_{IU\ ana2}$ $V_{IV\ ana2}$ $V_{IW\ ana2}$	4.7	4.9	5	V
Analog DC link voltage sensor output	load max 5 mA, @ 900 V	$V_{DC\ ana}$	6.4	6.5	6.6	V
Analog temperature sensor output unit 1 (NTC)	load max 5 mA, corresponds to $T_j = 125^\circ\text{C}$ at rated conditions	$V_{Theta\ NTC1}$		4.9		V
Analog temperature sensor output inverter section (NTC)	load max 5 mA, @ $T_{NTC} = 82^\circ\text{C}$	$V_{Theta\ NTC2}$		10		V

Notes

Over temperature shut down must be realized by customer.

prepared by: OW	date of publication: 2013-08-05
approved by: AR	revision: 2.0

Technical Information

PrimeSTACK™

6PS04512E43W39693



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System data

				min.	typ.	max.	
EMC robustness	according to IEC 61800-3 at named interfaces	power	V_{Burst}	2			kV
		control	V_{Burst}	1			kV
		aux (24V)	V_{surge}	1			kV
Storage temperature		T_{stor}		-40		85	°C
Operational ambient temperature	PCB, DC link capacitor, bus bar, excluding cooling medium	$T_{op\ amb}$		-25		55	°C
Cooling air velocity	PCB, DC link capacitor, bus bar, standard atmosphere	V_{air}		0.3			m/s
Humidity	no condensation	Rel. F		5		85	%
Vibration	according to IEC60721					5	m/s²
Shock	according to IEC60721					40	m/s²
Protection degree				IP00			
Pollution degree				2			
Dimensions	width x depth x height			215	280	120	mm
Weight					7.7		kg

Heatsink water cooled

				min.	typ.	max.	
Water flow	according to coolant specification from Infineon	$\Delta V/\Delta t$		10			dm³/min
Water pressure						8	bar
Water pressure drop		Δp			50		mbar
Coolant inlet temperature		T_{inlet}		-40		40	°C

Overview of optional components

	Unit 1	Inverter Section	Unit 3
Parallel interface board			
Optical interface board			
Voltage sensor		×	
Current sensor		×	
Temperature sensor		×	
Temperature simulation			
DC link capacitors			
Collector-emitter Active Clamping			

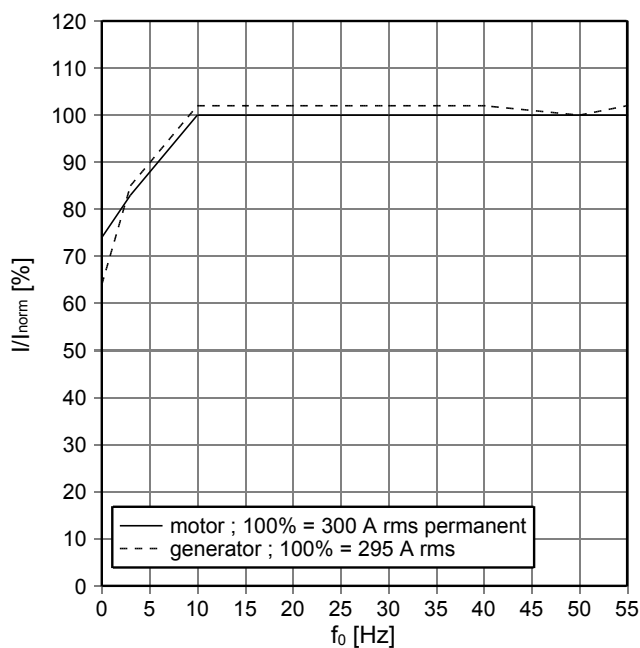
Notes

Setting of Active Clamping TVS-Diodes: $V_Z = 824\text{ V}$

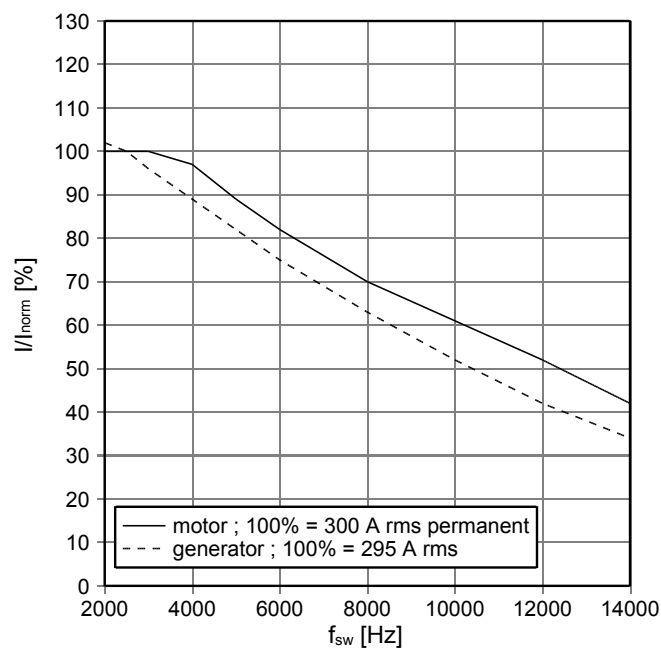
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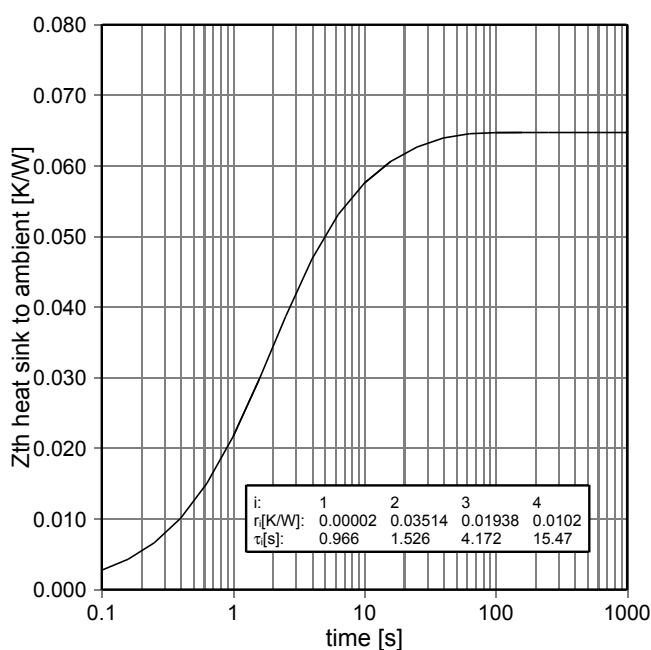
f_o - derating curve IGBT (motor), Diode (generator)
 $\cos(\phi) = \pm 0,85$
 $T_{cool\ medium} = 40^\circ C$



f_{sw} - derating curve IGBT (motor), Diode (generator)
 $\cos(\phi) = \pm 0,85$
 $T_{cool\ medium} = 40^\circ C$

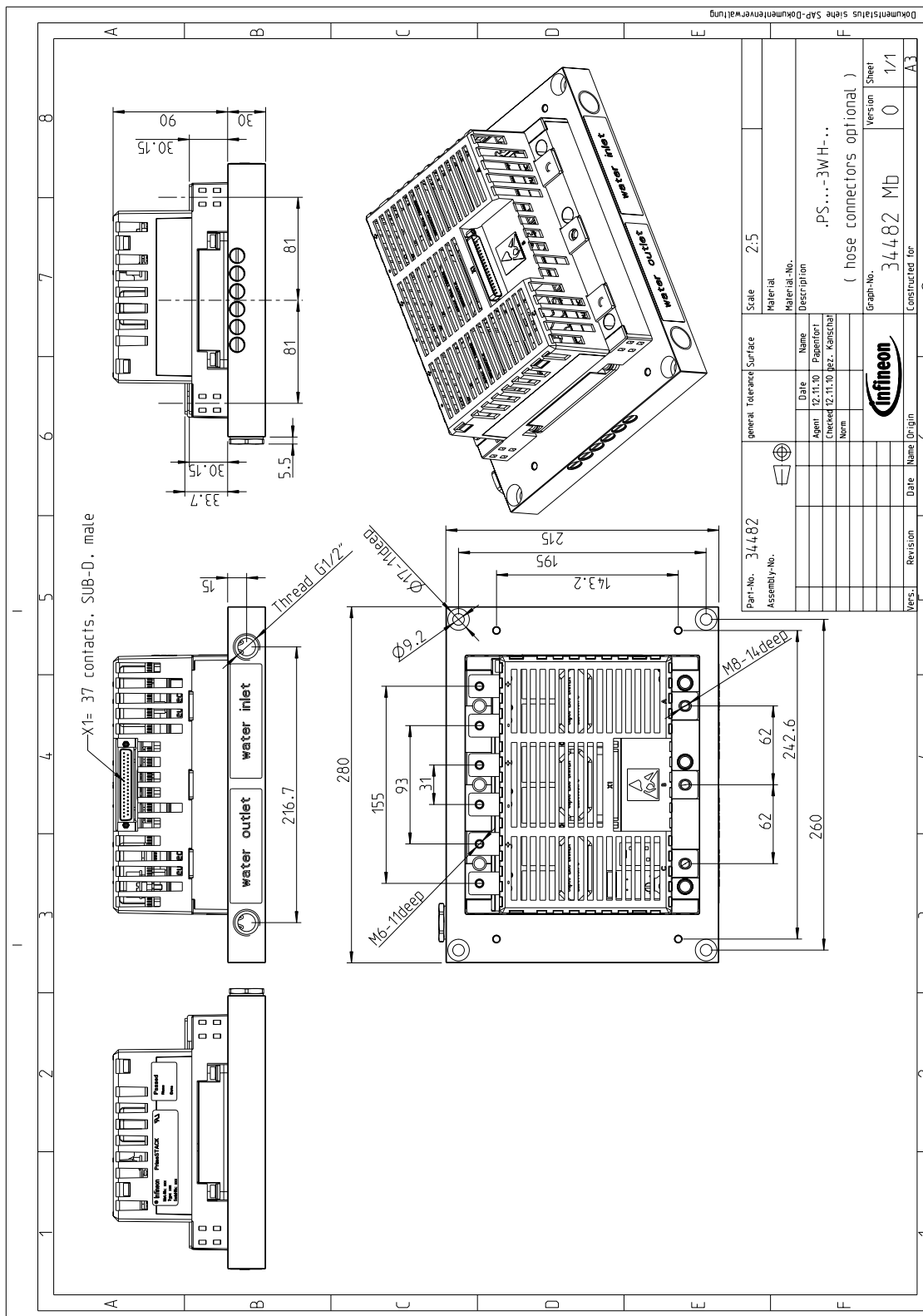


Z_{th} heat sink to ambient per switch
 $T_{cool\ medium} = 40^\circ C$



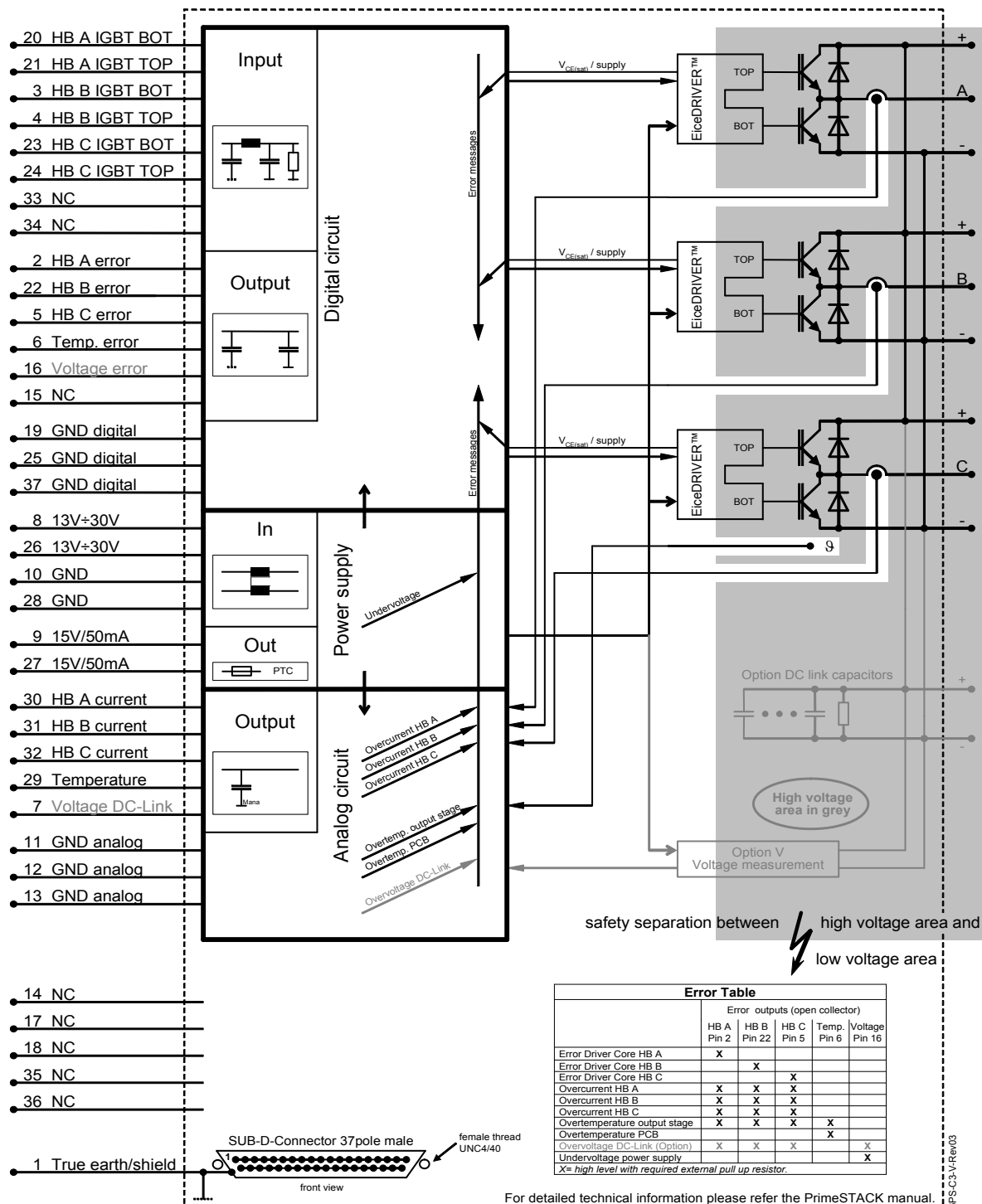
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Mechanical drawing



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Circuit diagram



Preliminary data

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This product data sheet is describing the characteristics of this product for which a warranty is granted. Any such warranty is granted exclusively pursuant the terms and conditions of the supply agreement. There will be no guarantee of any kind for the product and its characteristics.

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Should you intend to use the Product in aviation applications, in health or live endangering or life support applications, please notify. Please note, that for any such applications we urgently recommend

- to perform joint Risk and Quality Assessments;
- the conclusion of Quality Agreements;
- to establish joint measures of an ongoing product survey, and that we may make delivery depended on the realization of any such measures.

If and to the extent necessary, please forward equivalent notices to your customers.

Changes of this product data sheet are reserved.

Safety Instructions

Prior to installation and operation, all safety notices and warnings and all warning signs attached to the equipment have to be carefully read. Make sure that all warning signs remain in a legible condition and that missing or damaged signs are replaced. To installation and operation, all safety notices and warnings and all warning signs attached to the equipment have to be carefully read. Make sure that all warning signs remain in a legible condition and that missing or damaged signs are replaced.

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